

Lecture 6:

AI504 - Knowledge Representation

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1 $\mathcal{A}(\mathcal{RC})$: logic of “ a all q ”

Recal that this was enhanced by relative clauses (terms and verbs)

Today we will do soundness and completeness

2 Systems

We defined two proof systems

$A(RC)$

AXIOMS + BARBARA + DOWN

$$\frac{\text{all } x \text{ (r all } z) \quad \text{all } x \text{ } y}{\text{all } x \text{ (r all } y)} \text{DOWN DOWN}$$

Figure 1: Proof tree 1

$A(RC)'$

AXIOMS + BARBARA + ANTI

$$\frac{\text{all } x \text{ } y}{\text{all (r all } y) \text{ (r all } x)} \text{ANTI ANTI}$$

Figure 2: Proof tree 1

Note that both DOWN and ANTI, preserve semantic validity

2.1 Soundness

If $\Gamma \vdash \phi$, then $\Gamma \models \phi$

2.2 Completeness

If $\Gamma \models \phi$, then $\Gamma \vdash \phi$